

The Effect of Animated Thumbnails on Video Game Mods Metrics

Motivation

I am a video game mod developer actively working on creating mods for a game called **Garry's Mod**. Short for making modifications, *modding* can be defined as “nonprofessionals altering or adding to games” (Kretzschmar, 2019). In Garry's Mod (GMod) context, mods are usually called **addons** (not add-ons), and will be referred to as such throughout this report. For the past several years I have noticed a growing trend in the way addon representations are made in the **Steam Workshop**, the main modding platform provided within Steam by Valve Corporation.

Every Steam Workshop item has a **thumbnail**. Mod thumbnails can be of various dimensions and formats. Although specifically for the GMod Workshop the dimension is fixed to a square, there are many different image formats that can be used for thumbnails. A publishing utility provided within a community-created modding tool for GMod, *Crowbar*, allows for these following image file types:

1. **BMP** (Bitmap Image, *no compression*)
2. **GIF** (Graphical Interchange Format)
3. **JPEG** (Joint Photographic Experts Group)
4. **JPG** (*same as JPEG, exists due to Windows quirks*)
5. **PNG** (Portable Network Graphic, *lossless compression*)
6. **WMF** (Windows Metafile, *very old*)

You may have noticed that one of these file types has a unique feature not supported by the others. The GIF format is capable of displaying a recurring order of frames while staying under typical video sizes (Miltner, 2017). Basically it is an animated image format. The increased use of this format is what I have noticed as an emerging trend on the Workshop. Despite the very old file size limit for thumbnails of only a single megabyte (*which is very constrictive*), people have been making some very impressive looking animated thumbnails for their GMod addons.

This project is dedicated to potentially revealing the underlying reason behind the surge of GIF thumbnails (*will be referred as **Animated Thumbnails** from now on*) compared to the dominant **static thumbnail** formats of the past; in which static means every format other than GIF, on the given list above.

Data Source

Items in the Steam Workshop have built-in telemetry for tracking certain engagement. This data is provided by Steam automatically, meaning that data is presented directly on the Workshop page and indirectly under the “Item Stats” tab, which I will explain thoroughly later. We should first introduce the anatomy of a Workshop page to make things more clear. The page begins directly with some useful information:

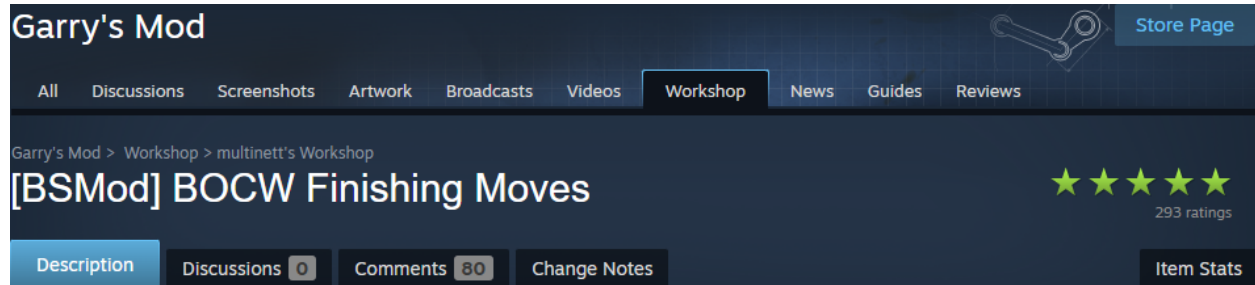


Figure 1. This is the top of the webpage -right below the header- of a Workshop item.

The very first thing to note is the game title on the top left. This page is part of the community hub of this game containing many places other than the Workshop. Right underneath that is the title of the Workshop item (or addon in our case). If the title is prefixed with a certain identifier in square brackets ([]), that usually means the mod is built on a certain codebase, in which it will not function without it. We start our discussion about the elements of interest with information on the right being stars and ratings.

Ratings are the total number of positive and negative ratings given by users on Steam. Figure 2 presents some of the buttons placed underneath the showcase images and thumbnail on the webpage. This is where users can like, dislike, award, favorite, share, report for violations and include the addon to one of their collections. The green **stars** in Figure 1 indicate some sort of relationship between the positive and negative ratings but it is not clearly known how the star counts are decided. The stars are not even displayed if there are too few ratings, in which it indicates “Not enough ratings” instead.

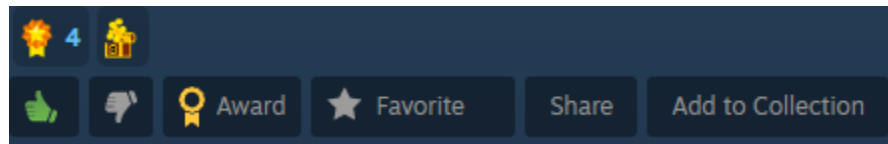


Figure 2. Some actions users are able to perform on a Workshop item. Awards are also displayed here.

From my experience, the page will not show any stars until 25 ratings, and will almost always start with 4 stars unless there is a significant amount of negative ratings. After around 250 ratings, the fifth star is awarded, given that there is an overwhelming amount of positive ratings. We can also see the number of **comments** up here.

The webpage continues with the subscribe button (which is equivalent to downloading the mod), the description, required items and the mod author. Right underneath the addon creator we observe some very important information.

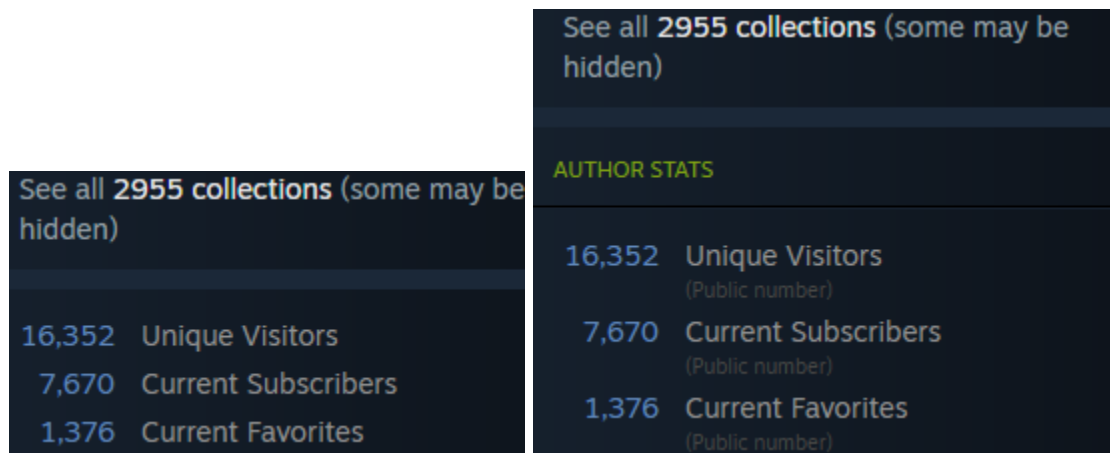


Figure 3. Stats shown beneath the mod author. The representation is different depending on whether you are the mod author or not, but the values are the same.

Here we can see the collection count and users are able to look up the collections that include this Workshop item by clicking the white text. **Collections** are mod-packs made by users containing addons that usually fit each other for a specific purpose or playstyle. But the really important data sits right underneath that; being unique visitors, current subscribers and current favorites.

Unique Visitors, according to the tooltip, “tracks when a new, logged in steam user visits your page”. **Current Subscribers** indicate the amount of users that have the addon downloaded at that specific moment. Similarly, **Current Favorites** tracks the amount of people that have favorited the addon at that time. Steam defines favoriting as “easily find this item again later”. The item will later show up in the “Your Favorites” tab while building a collection. There is not much more to talk about in this webpage.

Current Subscribers does not have direct relation to visitors due to users being able to subscribe to addons without visiting their pages, either via the workshop website or in-game. Due to this, some very old mods have significantly more subscribers than visitors.

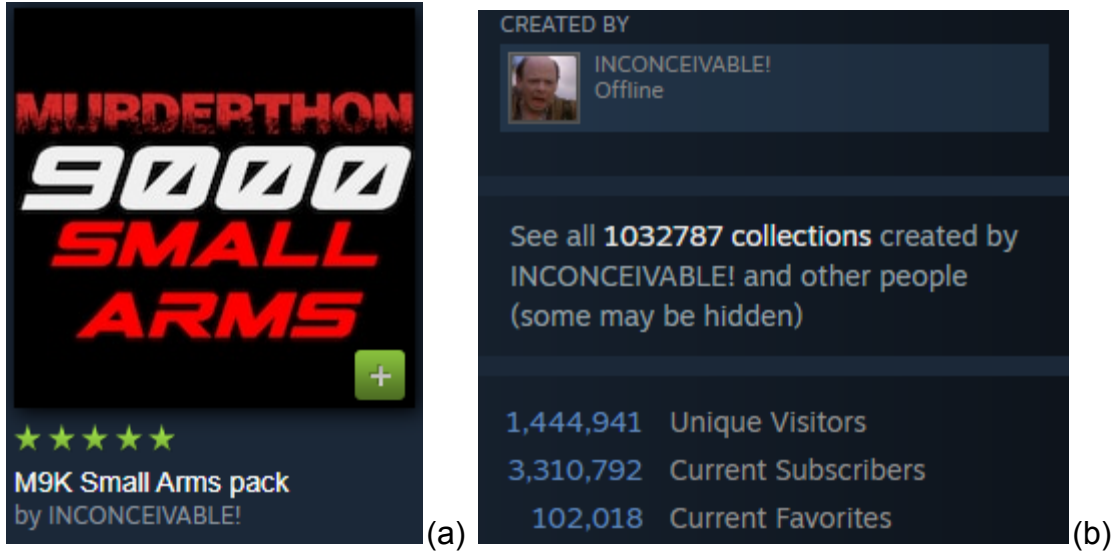


Figure 4. (a) Users can download a mod via hovering their cursor over the item and clicking the green plus button on the bottom right. The same action can also be done in Garry's Mod's built-in addon browser. (b) This may lead to subscribers exceeding the amount of visitors in some mods, which may seem like an error without this context.

The amount of data so far might have felt underwhelming. That is because I have not introduced you to the **Item Stats** tab yet in the Workshop page. This tab is only visible to mod authors and the webpage it leads to cannot be accessed by others. Figure 5 shows the outcome of trying to access the Item Stats tab on an item you do not own.

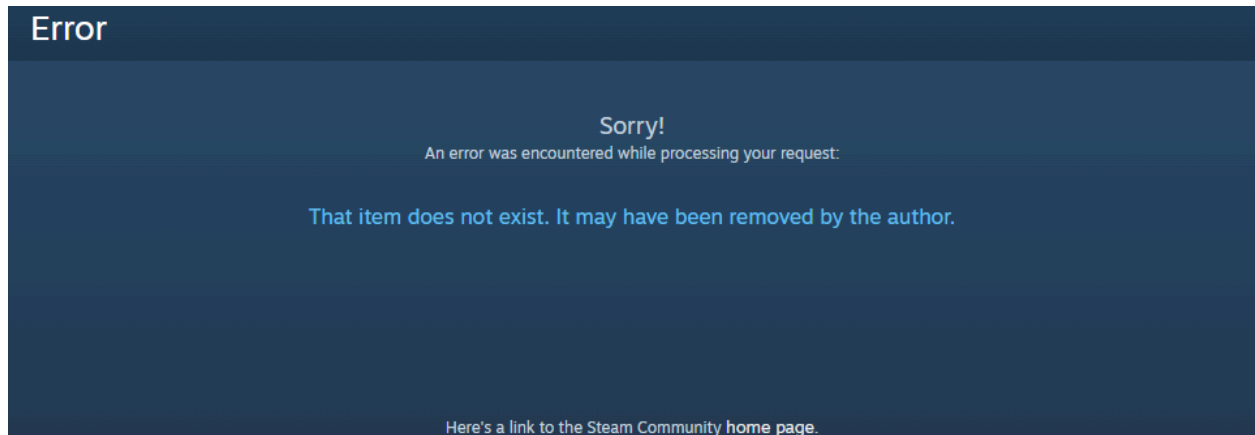


Figure 5. Error message that shows up when trying to access a private section in a Workshop item page. The error message is very misleading since it curiously displays the same error page when trying to access a removed or hidden Workshop item.

On the other hand, if you are the author of the Workshop page, you are greeted with this webpage instead, which is honestly the main inspiration for this project.



Figure 6. Example of an Item Stats page on a Workshop item.

There is a lot to unpack here so let's start with the familiar elements. The box with big numbers on the right is similar to the stats box displayed in the main page for the Workshop item, but instead of Current Favorites we have **Positive Ratings**, and we have an additional day counter displayed next to the Unique Visitors number.

On the left hand side, the same Unique Visitors number is present under VISITORS next to Total Unique. Beneath that is the SUBSCRIBERS field displaying the same green Current Subscribers number on the right. However right under that we come across a new metric called **Total Unique Subscribers**. It is defined as "a count of every user who had this item in their subscriptions, including users who later removed it". Next we have the FAVORITES field showing the same Current Favorites number we saw before and likewise to subscribers, we see a **Total Unique Favorites** entry, defined as "a count of every user who had this item in their favorites, including users who later removed it". Finally we have the RATINGS field hosting some useful information. We observe the total ratings statistic we saw before, and alongside that we can actually see the number of both Positive and **Negative Ratings**, with the percentage of each shown next to them.

We are not done with this page. Right below Figure 6 on the webpage, we can see a very useful and interesting graph, labelled **Cumulative Subscribers**.

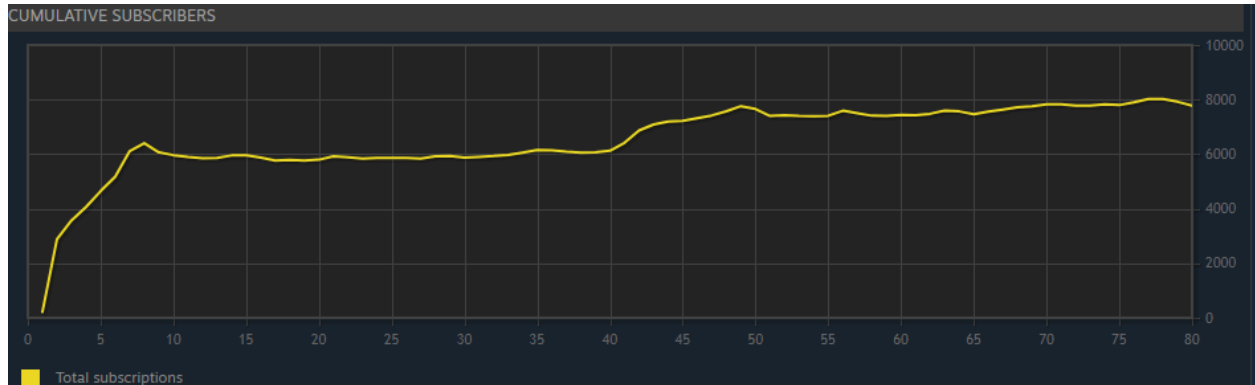


Figure 7. Cumulative Subscribers chart. This is an interactive graph giving us abilities like zooming into selection and inspecting individual data points.

The Workshop page apparently keeps track of the number of subscribers per given day after the mod has been released. It looks just like the chart of a stock, I guess if it was valued by how many subscribers the mod has. Anyway this is not the only feature Steam tracks and records on a daily basis. There is one final chart before the webpage comes to an end.

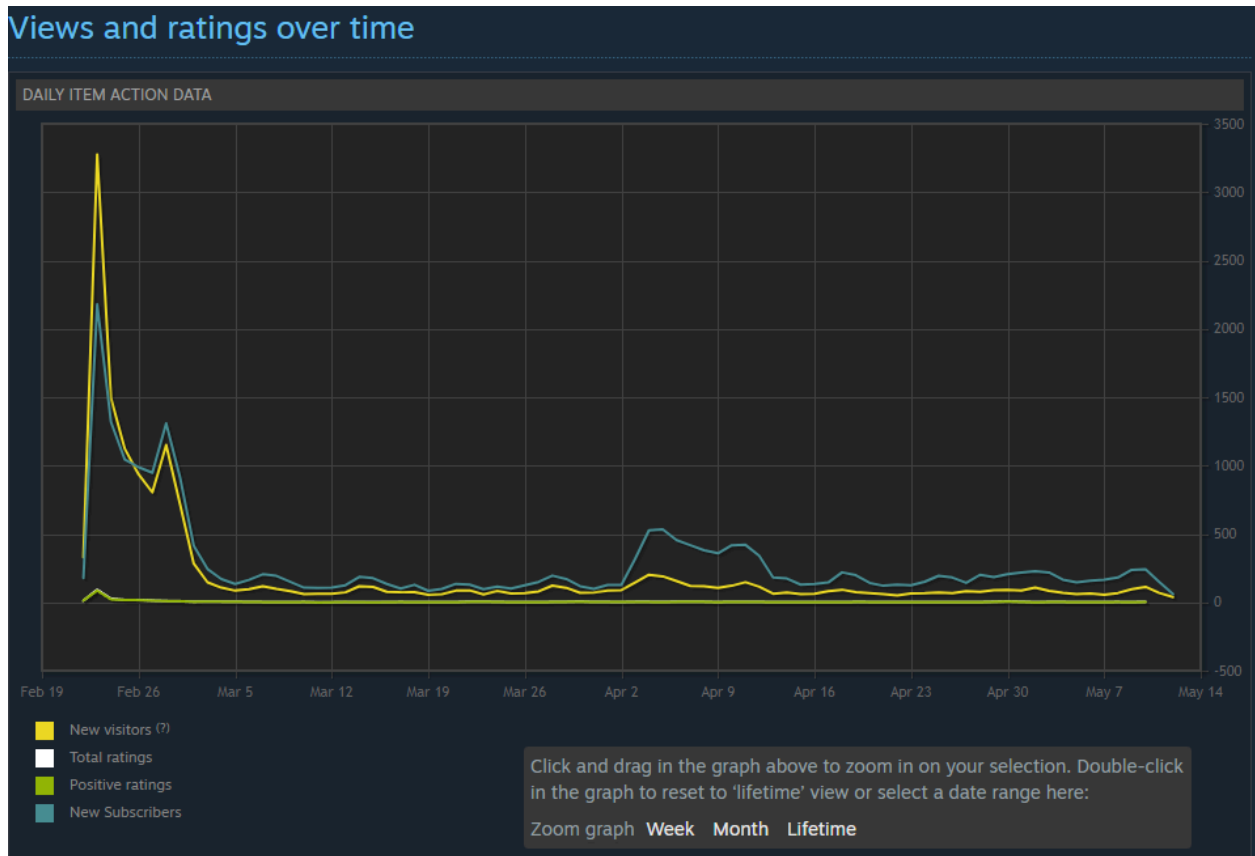


Figure 8. Views and ratings over time chart, labeled Daily Item Action Data.

We observe four features plotted as line charts. New visitors, shown by yellow, is daily data of how many Steam users clicked on the mod page for the first time. Total ratings shown as white is not visible on the chart for some reason. Positive ratings data is shown though, as green, indicating the amount of likes the mod got that day. The blue line depicts New subscribers acquired that day.

To summarize what we talked about so far, let's compile the data features together. We walked through the main tab and the Item Stats tab in a Workshop page. The main page is obviously visible to everyone but remember that the Item Stats tab is only visible to the authors. Here is the point where we should divide the features into two distinct categories: **Public Numbers** and **Private Numbers** (Table 1).

Table 1. Categorization of the various data features based on accessibility level.

Public Numbers	Private Numbers
Unique Visitors	Unique Subscribers
Current Subscribers	Unique Favorites
Current Favorites	Positive Ratings
Ratings & Stars	Negative Ratings
Comments	Cumulative Subscribers
Collections	Views and ratings over time

We should now elaborate on the reliability of all this data. Every element of Public Numbers appears to be very accurate, but this cannot be said for the other category. Within Private Numbers, Cumulative Subscribers and Views and ratings over time charts are not reliable all the time. In some Workshop items, the cumulative subscribers chart seemingly breaks, and shows negative amount subscribers, which obviously cannot happen. In the latter chart, sometimes one feature loses information or stops tracking for some time (*See Figures 9-12*).

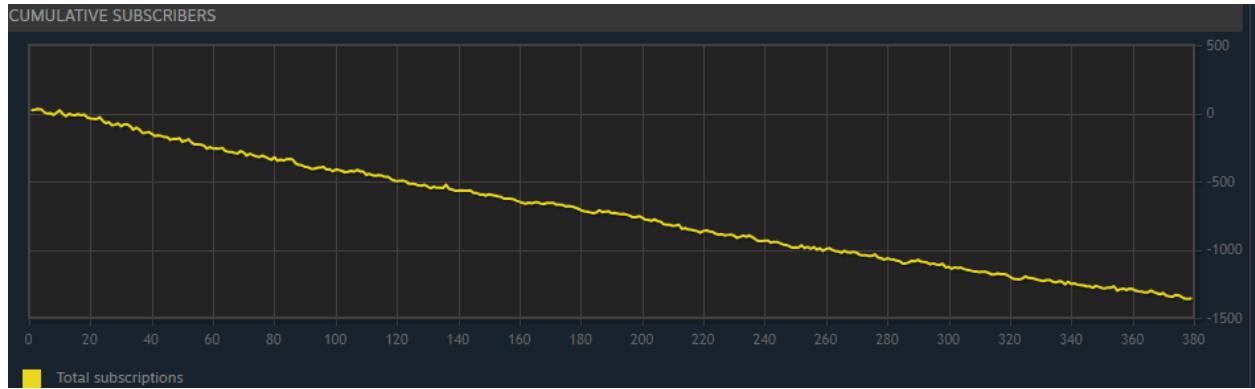


Figure 9. Example of a broken Cumulative Subscribers chart. Note the negative values on the right.

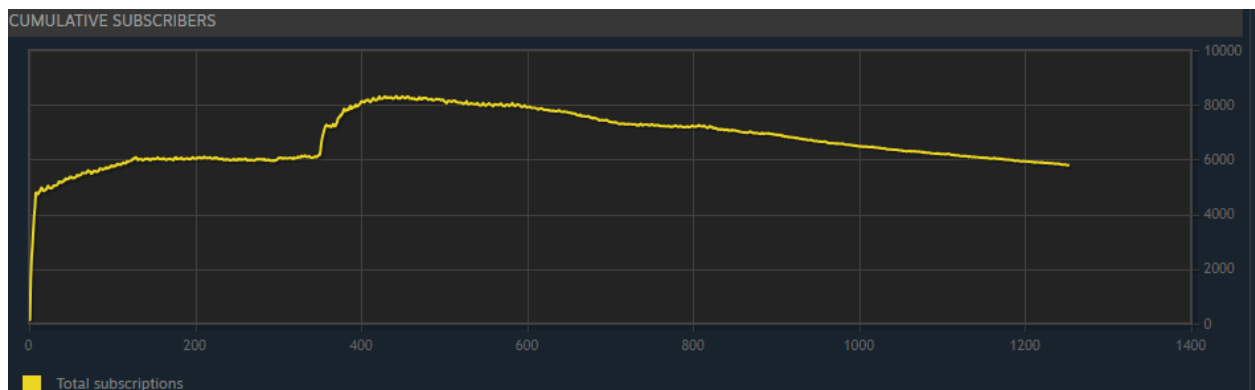


Figure 10. Cumulative Subscribers chart that has stopped tracking for reasons unknown.

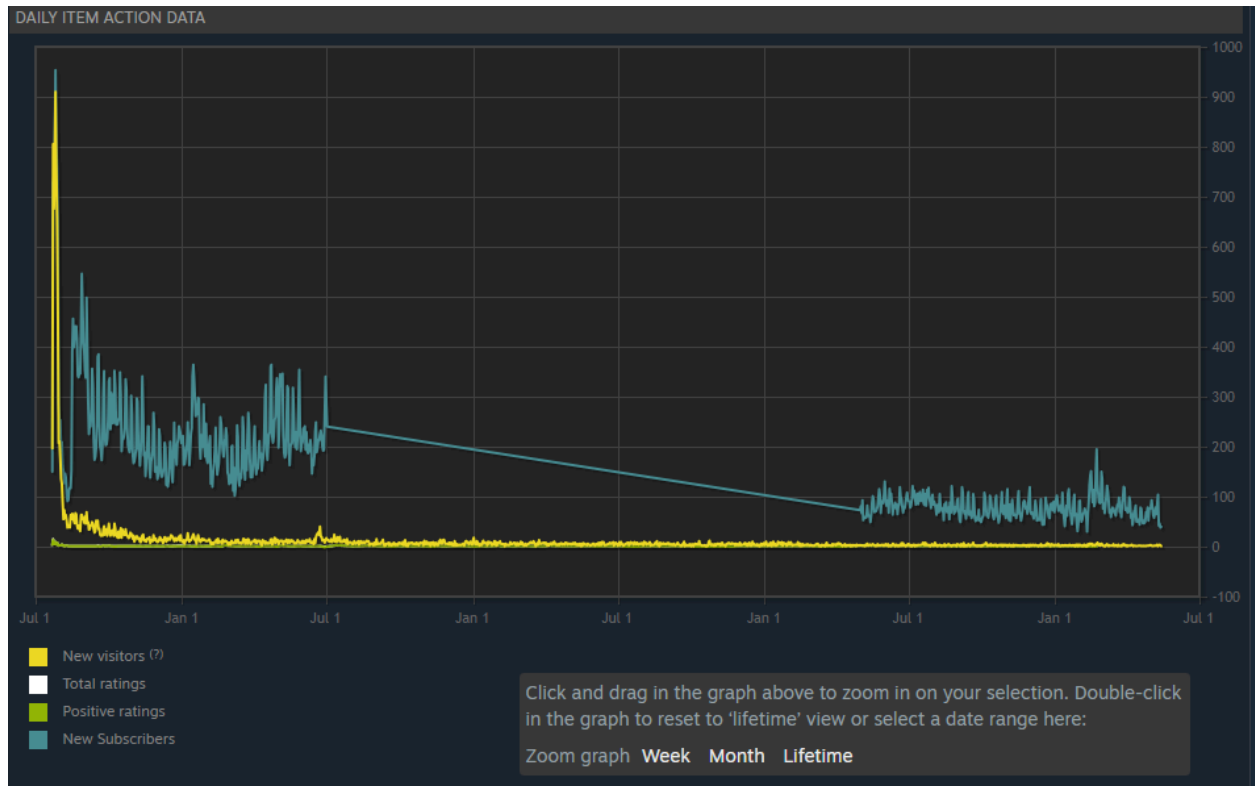


Figure 11. Daily item action data chart with nearly a whole year of missing data on the New Subscribers feature. This seems to happen very frequently.

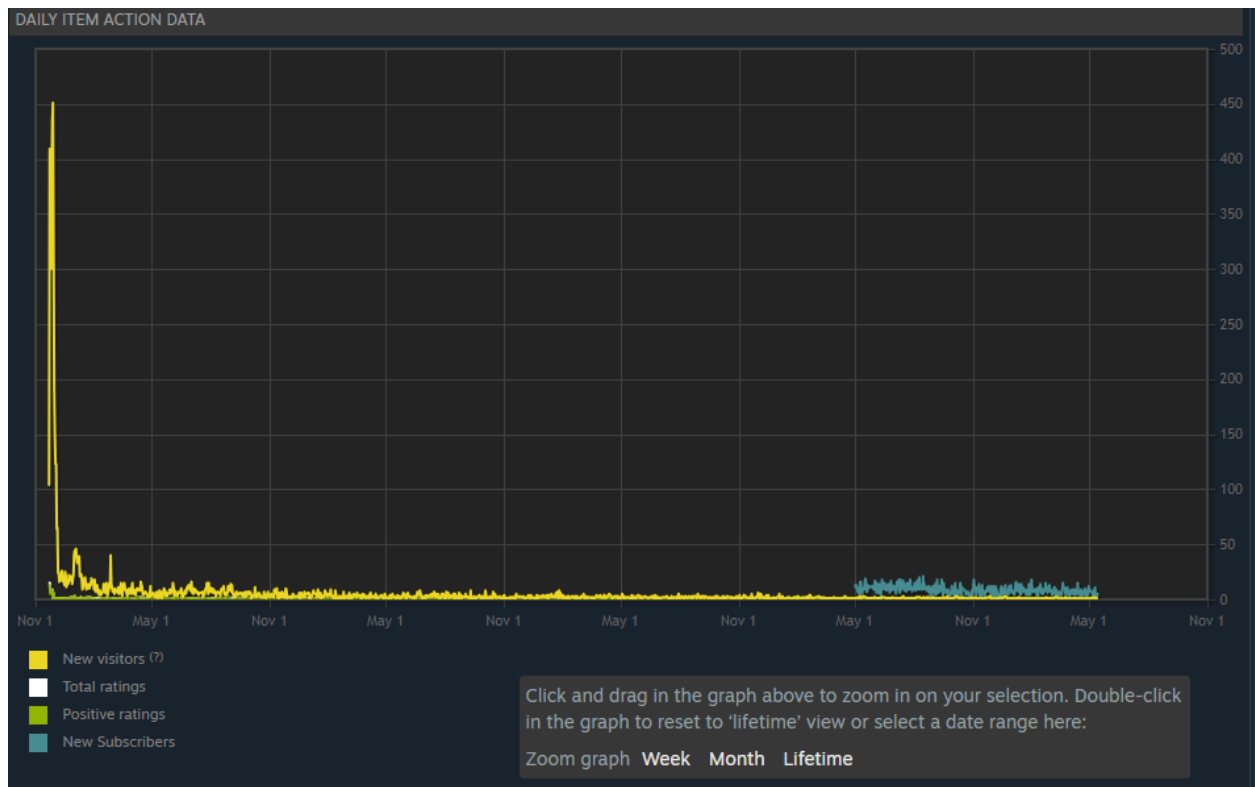


Figure 12. Daily item action data chart with lost New subscribers data.

The existence of Private Numbers is a problem for external research. Only the mod authors are able to conduct studies on their own addons with complete access to all information. An outside researcher has to either get Private Numbers data through contacting mod authors to be given access to the Item Stats tab, via adding them to the Workshop item contributors -which is beyond inconvenient and most if not all authors will not give this kind of access- or requesting for desired data from mod authors themselves, which is equally as hopeless for the researcher. Or the researcher could just not bother with Private Numbers at all and conduct limited research on the Public Numbers.

This brings us to the interesting part of data collection. At the beginning I mentioned that I am a mod developer myself. At the time of writing this, I have posted or been a contributor to almost 150 addons on the Workshop. This means I am in an interesting position where I am very much capable of conducting in depth research on the mods I have made or contributed, however this kind of focus on personal Workshop items may fail to give good answers to a generalized research question I am seeking to find an answer to.

Due to this, I have decided to create and conduct work on not one but two datasets. The **Public Dataset** is a judgment sample made of 100 addons from ten different addon types, in which five have static thumbnails and five have animated thumbnails. For reference, the addon types in the Garry's Mod Workshop are:

- ❖ Gamemode
- ❖ Map
- ❖ Weapon
- ❖ Vehicle
- ❖ NPC
- ❖ Tool
- ❖ Entity
- ❖ Effects
- ❖ Model
- ❖ Server content

Each addon type was given equal weight for generalization and enriching purposes. For addon selection, we came up with a criteria for fairness, which are as follows:

1. The addon must be released in 2025.
2. The addon must not be an obvious dependency or a library.
3. The addon must have over 3 stars.

On top of the public dataset, due to reasons I mentioned regarding the existence of Private Numbers, I have also put together a **Private Dataset**, consisting of 17 animated and 17 static thumbnail mods made by myself. The same selection criteria applies except the 2025 limitation. The dataset spreadsheets can be accessed using links provided under the *Appendixes* section or in the GitHub repository under the Datasets folder.

Data Analysis

Our aim is to figure out whether addons with animated thumbnails perform differently (or perhaps better) compared to mods that have been published with static thumbnails. For this we made heavy use of **hypothesis testing**, more specifically, **independent samples t-test**. On both datasets, we conducted hypothesis tests on every available feature in Public and Private Numbers.

After that we applied **linear regression** on the Private dataset, specifically on the cumulative subscribers data. For both animated and static thumbnails we averaged the performance of the first 60 days in terms of subscriber count in order to model the increase and retention of mod users. Addons with previously mentioned “broken” cumulative subscriber data were discarded in this process.

Findings

We begin our discussion with results obtained from the public dataset. Here the hypothesis testing on Public Numbers were conducted, but not on the Private Numbers for reasons discussed in detail previously. We care about whether animated thumbnails have a significant effect over the metrics that are tracked by the Steam Workshop. Our significance level is set to a standard 95%, meaning that the p-values we obtain through randomization dotplots should remain under 0.05 in order for us to reject the null hypotheses.

The public dataset turned out to be of not much value. On every public number, nothing of significance was found. Unique Visitors –the metric we expect to be the most affected since thumbnail type should have direct influence on users clicking the mod– was higher on static thumbnails. Subscriber counts were relatively similar and the amount of favorites were insignificant. Collections, comments and ratings were also not significantly better or worse.

Things did change however when we continued to the private dataset, collected from my own mods. Remember that this dataset contains valuable Private Numbers. Unique

Visitors were significantly higher on animated thumbnails with a p-value of 0.03. Similarly there were also more subscribers giving a p-value of 0.025. Collections were also significantly higher for animated thumbnails ($p = 0.032$). There were also more stars ($p = 0.044$) and positive ratings ($p = 0.041$). However, favorites (0.088), comments (0.136), negative ratings (0.804), total unique subscribers (0.242) and total unique favorites (0.245) were statistically insignificant (Table 2).

Table 2. Significance level of metrics after hypothesis testing in the private dataset.

Significant Numbers	Insignificant Numbers
Unique Visitors	Current Favorites
Current Subscribers	Comments
Collections	<i>Negative Ratings</i>
Stars	<i>Unique Subscribers</i>
<i>Positive Ratings</i>	<i>Unique Favorites</i>

Unique Visitors was expected to be significantly higher however subscribers and collections were a surprise. Stars significance most likely came from the directly related positive ratings. Most private numbers (shown in italics in Table 2) turned out to be insignificant.

ML conducted on the private dataset revealed that animated thumbnails have gained and retained more subscribers than static thumbnails, although the retention and gradual increase of subscribers later on look similar (Figure 13).

All in all, the outcome seems to be inconclusive. The public dataset results do not agree with the private dataset findings. We could have been more confident regarding the effectiveness of animated thumbnails if both datasets were pointing towards the same direction. In the next section we discuss the potential limitations and reasons behind why we could not come up with a clear and concise answer.

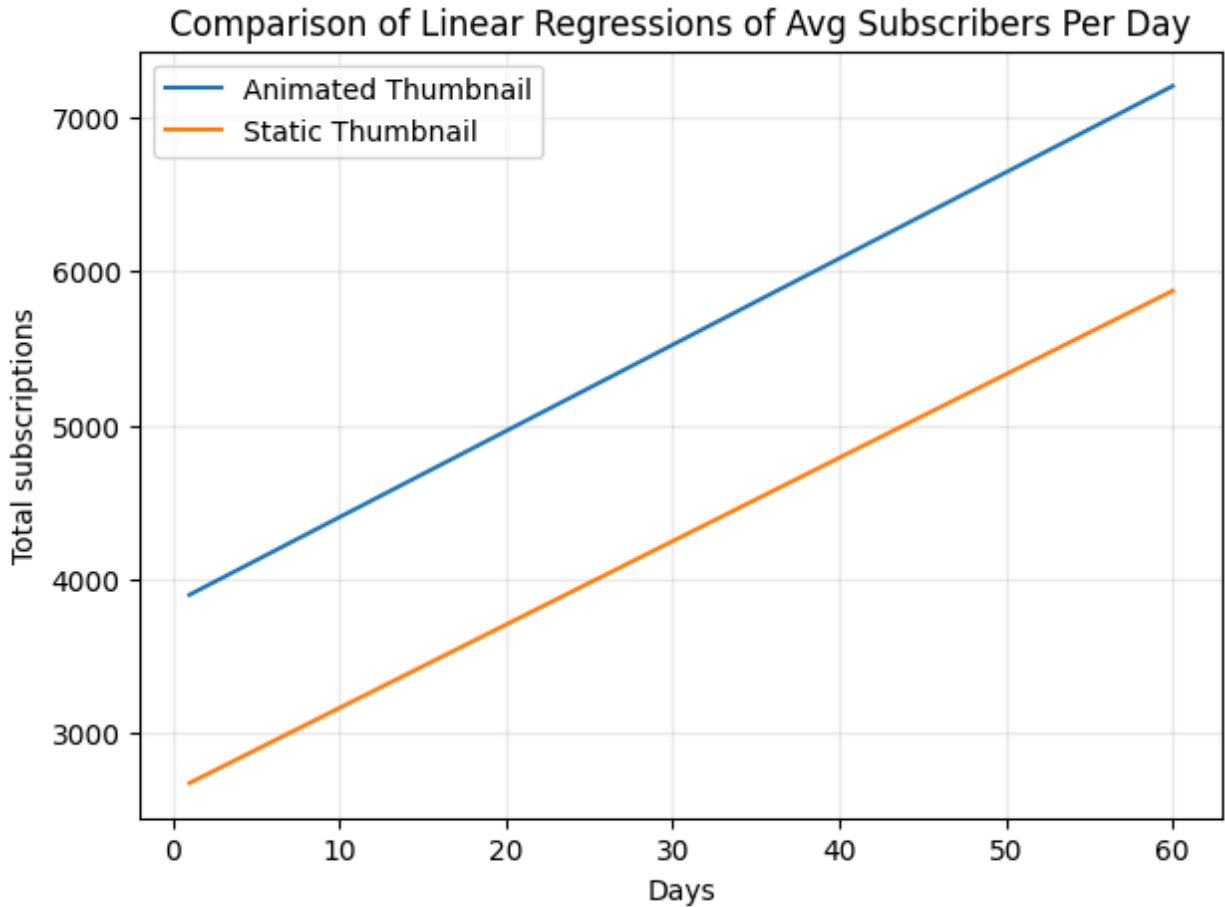


Figure 13. Linear regression comparison of cumulative subscribers of the two thumbnail types.

Limitations and Future Work

Likely the biggest limitation of this study is the absence of web scraping. Both datasets were made with manual data input which limited sample size, caused inefficient use of time and introduced potential human error. A proper web scraping tool could not be developed due to lack of experience regarding that and time constraints. Much broader analysis could have been done or findings may have been found if the public dataset, via automation, consisted of potentially thousands of addons. In addition to that, findings that came from the Private dataset may not be representative of the general Workshop environment as mentioned before.

The “Views and ratings over time” chart was also never used due to reliability concerns. Future work can potentially find a way to use this chart and evaluate the rest of the numbers in better and improved ways. For ML, the linear regression model does not fit

the clearly non-linear cumulative subscribers scatterplot well, and leaves much room for multinomial regression.

We believe that the rise of animated thumbnails did not come from nowhere. There has to be a definitive reason for and outcome from deciding to go for an animated format for item preview images. It could be said that animated thumbnails generally did better on my end, but an answer that applies to the entire GMod Workshop can only be provided by future work in this very unexplored field.

References

Kretzschmar, M., & Stanfill, M. (2019). Mods as lightning rods: A typology of video game mods, intellectual property, and social benefit/harm. *Social & legal studies*, 28(4), 517-536.

Miltner, K. M., & Highfield, T. (2017). Never gonna GIF you up: Analyzing the cultural significance of the animated GIF. *Social media+ society*, 3(3), 2056305117725223.

Appendixes

Datasets were prepared using Google Sheets, here are the links to the original spreadsheets for transparency:

[Public Dataset](#)

[Private Dataset](#)

AI Disclosure

No generative AI and/or LLM tools were used in any part of the project or this report.